

CMS Pediatrics Form 1 - Answers & Explanations

1. E) Trimethoprim-Sulfamethoxazole
 - TMP-SMX inhibits enzymes in the DNA synthesis pathway → megaloblastic anemia, leukopenia, granulocytopenia
 - Avoided with co-administration of leucovorin (folic acid)
2. D) Elimination of lead from the child's home
 - Lead impairs heme synthesis → microcytic anemia
3. F) Shallow, poorly developed acetabulum
 - Barlow & Ortolani maneuver to screen for developmental dysplasia of the hip
 - "click" = femoral head dislocating posteriorly with reduction of the femoral head into the acetabulum
 - More common in firstborn girls born in breech position
4. D) Pathologic fracture of the humerus.
 - Clear pathological fracture of the humerus on x-ray
 - Ewing sarcoma → 'onion skinning'
5. F) Major depressive disorder
 - This patient has ≥ 5 of the 9 diagnostic symptoms of SIG-E-CAPS for ≥ 2 weeks
6. A) Closure of the ductus arteriosus
 - Cyanotic newborn with no improvement on O₂ therapy = cardiac cause
 - This patient likely has Coarctation of the aorta that is dependent upon the patent ductus arteriosus (PDA is distal to the coarctation)
 - PDA closure → new onset CHF (severe hypotension, poor perfusion, tachypnea, hepatomegaly, unexplained metabolic acidosis)
7. E) Thick and thin blood smears
 - This patient went to a chloroquine resistant area → malaria
8. E) Myotonic dystrophy
 - Usually presents with facial weakness, hand grip myotonia, dysphagia
 - Arises from autosomal dominant expansion of CTG trinucleotide
9. A) Cerebral edema
 - Decreasing plasma osmolarity too quickly → fluid shifts from ECF to ICF → $\uparrow$ ICP
 - $\uparrow$ ICP → absence of venous pulsations
10. A) Reduced-calorie diet
 - Lifestyle (diet and exercise) modification is first-line (exercise should be aerobic; weight training = anaerobic exercise)
11. D) Staphylococcus aureus
 - Fungal infection → dry skin that cracked and allowed bacteria from the skin to get in → cellulitis secondary to staph infection → fever & pain.
12. F) Legg-Calvé-Perthes disease
 - Commonly presents between 5-7 years old with insidious onset of hip pain
 - Slipped capital femoral epiphysis commonly presents in obese pre-teens
13. D) Cruciate incision of the hymen
 - Adolescent girl at usually age of menarche presenting with amenorrhea, cyclic abdominal/pelvic pain and hematocolpos = Imperforate Hymen
 - Hematocolpos (blood-filled vagina) → bulging, blue hymen membrane
14. B) Hepatitis A Virus: immune, Hepatitis B Virus: not immune
 - Hepatitis A antibody = immune
 - Anti-HBc (HBV core antibody); IgM indicates acute/recent infection, IgG indicates prior exposure or chronic infection
 - HBsAg (HBV surface antigen) indicates hepatitis B infection
 - Lack of Anti-HBs (HBV surface antibody) indicates no immunity

15. D) Pulmonary artery hypertension
- Eisenmenger's syndrome: chronic left-to-right shunting → pulmonary artery HTN → shunt reversal → cyanosis and polycythemia
16. E) Hypothermia
- J wave = positive deflection occurring at the junction between the QRS complex and the ST segment (J wave = Osborn wave)
 - J wave are suggestive of hypothermia (but are not pathognomonic)
 - Height of J wave is proportional to the degree of hypothermia
17. C) Nasal polyps
- Chronic URIs and steatorrhea are consistent with cystic fibrosis (CF)
 - ~10-40% of CF patients have nasal polyps (due to chronic inflammation)
18. E) Physiologic pubertal development
- Gynecomastia occurs in ~33% of pubescent boys; goes away with development
19. C) A 6-week course of oral AZT within 12 hours after delivery
- * guidelines on this are in flux - unlikely to be tested*
 - If mother is HIV ⊕ then baby gets 6 weeks of AZT within 6 hours of birth
 - Whether to add on other antiretrovirals after that is based on mom's viral load and other risk factors
20. B) Intravenous cefotaxime
- High leukocyte count (indicates infection) warrants antibiotics • She already takes penicillin prophylaxis (the bug she has is probably resistant)
 - 3rd generation cephalosporins treat possible meningitis
21. E) Impaired humoral immunity
- Common variable immunodeficiency (CVID) → defect in B-cell differentiation → ↓ plasma cells and immunoglobulins → ↑ sinopulmonary infections
 - Combine immunodeficiency (eg, SCID) would have evidence of recurrent/severe viral and fungal infections (not just bacterial infections)
22. D) Serial blood cultures
- Fever + new systolic murmur (mitral regurgitation) = Endocarditis
- 23) Exocrine pancreatic insufficiency
- Failure to thrive, steatorrhea, meconium ileus, ↓ soft-tissue mass (< 5th percentile for weight) = Cystic fibrosis (CF)
 - Pancreatic insufficiency → malabsorption with steatorrhea → fat-soluble vitamin deficiencies (A, D, E, K)
24. B) Adams-Stokes attack
- P waves, no QRS = AV block • High grade AV block (ie, complete or Mobitz type II) = Stokes-Adams attack
 - Tx: Pacemaker
25. P) Sinusitis
- >10 day history of fever, headache, and nasal discharge
 - Post-nasal drip → erythematous posterior pharyngeal wall
26. B) Bacterial pneumonia
- Decreased breath sounds, right lower lobe density (consolidation), abdominal pain (referred pain secondary to diaphragmatic irritation from basal pneumonia or from overuse of abdominal muscles secondary to persistent cough)
27. C) Osteoporosis
- Modifiable risk factors for osteoporosis: ↓ physical activity, low body weight, excessive ethanol or tobacco use, premature menopause, glucocorticoid use
 - This patient is headed toward the female athlete triad: low BMI, irregular periods, and fractures
28. C) Posterior urethral valves
- Most common cause of urinary tract obstruction in newborn boys
 - Vesicoureteral reflux is a common cause of recurrent UTI in kids

- 29) Pyloric stenosis
- Forceful, projectile, non-bilious vomiting
 - Dehydration & weight loss
 - Dx: abdominal U/S
 - Tx: Pyloromyotomy
- 30) T lymphocytes
- T-cell deficit → fungal/viral infections
31. C) Impetigo contagiosa
- Honey-crusted lesions ≈ Impetigo
 - Unlikely to be herpes simplex since the lips are spared
32. D) Shigellosis
- Fever, campy abdominal pain, dysentery = Shigella
 - Seizures = common neurologic complication of shigella; associated with fever > 39 °C (102.2 °F)
 - Ekiri syndrome = lethal toxic encephalopathy due to shigellosis
 - Characterized by rapid development of seizures and coma in patients with high fever & dysentery
33. D) Silver stain of bronchoalveolar fluid
- Pneumocystis pneumonia (caused by the yeast-like fungus *Pneumocystis jirovecii*)
 - Mother had history of IV drug use
 - Baby's frequent thrush and failure to thrive (FTT) = immunocompromised
34. C) Calcium
- Seizures & prolonged QT = symptoms of hypocalcemia
 - Infants of diabetic mothers often have decreased magnesium and calcium
 - Infants have difficulty absorbing calcium due to suppressed PTH (ketonuria → ↓ magnesium → suppression of PTH → ↓ calcium)
35. D) Delayed gross motor, normal fine motor, normal language development
- Child should be sitting up by himself after 6 months (this patient is 9 months old and is still unable)
36. E) Hyperglycemia
- Pseudo-hyponatremia from hyperglycemia
 - Hyperglycemia causes osmotic shifts of water from the ICF to the ECF → dilution hyponatremia
 - Hyperglycemia → ↑ diuresis → osmotic diuresis → hyponatremia
37. A) Serum lipid studies
- AAP recommends universal lipid studies for kids age 9-11 (pre-puberty), and again at age 17-21 (post-puberty)
 - "heart disease" in father and grandfather ≈ hypercholesterolemia
38. C) Inguinal hernia
- "milked" = reducible hernia
39. C) Measurement of serum β-hCG concentration
- Always rule out pregnancy first, even if she hasn't begun menstruation yet (tanner stage is highly indicative of impending menstruation)
 - X-ray & CT = radiation exposure = dangerous for potential fetus
40. C) Bronchoscopy
- Foreign body → air-trapping → hyperinflation → hyper-resonance & ↓ breath sounds
 - Bronchoscopy for removal of foreign body
41. E) Venous hum
- Innocent, benign murmur in children
 - Turbulent flow of venous return in the jugular veins and SVC; occurs during systole and diastole
 - Accentuated or eliminated with head position; disappears when supine or with digital compression of the jugular vein
42. A) Aplastic anemia
- Aplastic anemia = pancytopenia (↓ RBC, ↓ WBC, ↓ Platelets)
 - Aplastic crisis = ↓ Erythrocytes

43. E) Herpes simplex virus
- Temporal lobe hyper/hypo-densities = HSV
 - Herpes encephalopathy = ↑ RBC, normal glucose, ↑/normal protein, ↑ lymphocytes
44. D) Voiding cystourethrography
- Question is out of date - Should be 'Observation for recurrent symptoms'
 - Does not meet indications for VCUG (voiding cystourethrography)
 - Posterior urethral valves are the most common cause of UTI in a baby boy
 - Voiding cystourethrography checks the posterior urethral valves
 - However, cystourethrogram is generally not indicated in first time UTIs unless U/S shows hydronephrosis, scarring or the patient is < 1 month old
45. F) Tuberous sclerosis
- Skin hypopigmentation = ash-leaf spots
 - Periventricular masses = subependymal astrocytomas
46. C) Lymphocytic infiltration
- Hashimoto thyroiditis → lymphocytic infiltrate (autoimmune process)
 - Lymphoid aggregates with germinal centers
 - MCC of hypothyroidism in iodine-sufficient regions
47. D) Mixed metabolic acidosis and respiratory alkalosis
- Salicylate toxicity (early) stimulates medulla respiratory centres → hyperventilation → respiratory alkalosis
 - Salicylate toxicity (late) decreases cellular respiration via uncoupling oxidative phosphorylation → anaerobic metabolism → metabolic acidosis
 - This child was brought to the ED 20 minutes after her mother found her, not 20 minutes after ingestion of the aspirin...
48. B) Cooking meats completely
- Humans and animals are reservoirs for Salmonella species (except Salmonella typhi)
 - Humans are the only reservoir for S. typhi
 - Poultry (BBQ chicken), eggs, pets, and turtles are common sources
 - Non-typhoidal Salmonella → gastroenteritis
49. E) Strict hand-washing techniques at the day-care center
- Rotavirus is transmitted through the fecal/oral route via contact with contaminated hands, surfaces and objects.
50. D) Phototherapy
- Unconjugated hyperbilirubinemia from birth trauma (cephalohematoma)
 - This baby has been breast-feeding and has normal stool & urine output
 - If the child had a positive Coombs test and signs of acute bilirubin encephalopathy (eg, high-pitched cry, lethargy, poor feeding, and hypotonia), they should be treated with exchange transfusion as this is a medical emergency
 - Mild elevation (>10 mg/dL) Rx: phototherapy
 - Severe elevation (>20-25 mg/dL) Rx: exchange transfusions